

BRAIN TUMOR CLASSIFICATION: A DEEP LEARNING APPROACH

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ABSTRACT

The following is a project proposal for APS360: Introduction to Deep Learning.

—Total Pages: 6

1 INTRODUCTION

Over 300,000 brain tumours are diagnosed annually worldwide (Cohen-Gadol, 2024). The fast-paced nature of this disease makes early detection vital for a patient's survival. Typically detection is done by acquiring medical images (i.e. MRI scans) which are analyzed by experts for tumors. This task lends itself to deep learning since a model can be trained to detect brain tumours in images based on already classified images by experts. This model can be used as a tool in the analysis of brain scans for verification of the expert's findings or in situations where an expert is not readily accessible. Thus the project proposed is a deep learning model capable of detecting tumors in brain scan images.

2 BACKGROUND AND RELATED WORK

- 1. Automated Brain Tumor Detection and Classification Using Convolutional Neural Networks:** This is an IEEE study which demonstrates an automated method for diagnosing and classifying brain tumors using Convolutional Neural Networks (CNNs) as they are exceptionally good at processing high-resolution medical images (M. Nagraju, Apr. 2024). The researchers used high-resolution MRI scans of the brain taken from several different datasets which emphasize the importance of quality images when it comes to training the model to achieve near-perfect accuracy. When it comes to the architecture, the researchers integrated attention mechanisms as well as multi-scale feature fusion to increase the model's sensitivity to different tumor features. The model achieves a 99.76% accuracy in the identification as well as classification of tumors in MRI scans, surpassing the tumor classification experts.
- 2. Deep Learning Suite for 3D Image Segmentation** DeepMedic is an open-source deep learning framework developed to process medical images, especially brain lesion segmentation. It uses a 3D deep convolutional neural network coupled with a 3D fully connected conditional field (deepmedic.org, 2024). This allows for the processing of 3D medical images such as MRI and CT scans. This model has implemented multi-scale input architecture, thus it can process images at different scales, allowing it to detect features varying in size, improving its accuracy.
- 3. Enhancing brain tumor detection in MRI images through explainable AI using Grad-CAM with Resnet 50** This study by BMC Medical Imaging used Resnet 50 which

is a CNN architecture known for its ability to extract small details from an image (M. M. Musthafa, May 2024). To increase the transparency of the model, making the tumor identification process more interpretable, the model uses Grad-CAM (Gradient-weighted Class Activation Mapping) which illustrates which part of the image the model focuses on to make its decisions. Data augmentation was also used which is the process of modifying the images in already existing datasets to make the model generalize and learn better. The model achieved an accuracy of 98.52% as well as a high F1 score.

4. **A robust grey wolf-based deep learning for brain tumour detection in MR images**

This study employed contrast enhancement and skull stripping to remove the excess noise, thus allowing the model to focus on the important features of the MRI scans (A. Geetha, 2018). They also use the Fuzzy C-Means Clustering (FCM) algorithm to isolate the regions of interest for tumor detection. Furthermore the Gray Co-occurrence Matrix (GLCM) and Gray-Level Run-Length Matrix (GRLM) enable the model to extract specific features from the MRI scans. This study takes a unique approach of using the Deep Belief Network (DBN), optimized using Grey Wolf Optimization (GWO) to identify brain tumors as opposed to CNNs used by the other studies mentioned thus far.

5. **Brain Tumor Detection Based on Deep Learning Approaches and Magnetic Resonance Imaging**

This study fine-tuned a YOLOv7 deep learning model using the transfer learning technique to improve its performance. In addition, feature enhancements such as Convolutional Block Attention Module (CBAM) and Spatial Pyramid Pooling Fast+ (SPPF+) were added (A. B. Abdusalomov, 2023). The study concluded that the model had a 99.5% accuracy in detecting glioma, meningioma, and pituitary brain tumors.

3 DATA PROCESSING

The project utilizes two main datasets, a 3064 image dataset from Figshare and a 3000 image dataset from Kaggle.

3.1 OVERVIEW OF THE DATASETS

The figshare dataset contains 3,064 brain tumour images collected between 2005 and 2010 from Nanfang Hospital, Guangzhou, China, and General Hospital, Tianjin Medical University, China (Cheng, 2017). Since this dataset only has tumorous images, an additional dataset from Kaggle, the Br35H dataset, will be used, which contains 3,000 images, with 1,500 tumorous and non-tumorous images each (Hamada, 2021).

3.2 CHALLENGES

There are several difficulties in processing the datasets:

- Different image sizes: Crop the images with extra dimensions and pad the images that are too small.
- Different image formats: One of the datasets provides the images in a Matlab variable and the other stores the images as JPG files. To ensure consistency, all of the images need to be converted to the same format.
- Relabeling: Most of the images are not labeled in an accessible format or the labels are not suited to our needs. This will require a relabeling script to allow for uniform labeling for all datasets.
- Insufficient Data/Uneven Distribution: There are 3 times more tumorous images than non-tumorous images. In addition, 6000 images may not be sufficient to train the model. This can be resolved by data augmentation. We can generate new images by rotating the images and reflecting them horizontally and vertically.
- Contrast Mismatch: One of the datasets is contrast-enhanced, while the other is not. This can lead to inconsistent feature extraction for the CNN which can result in a less accurate model.

4 ARCHITECTURE

Since this project requires the model to train on thousands of images, the most viable option for the architecture is to use a CNN. This is because CNNs allow for the use of filters to extract key features of a tumor such as edges, size, etc. The extracted features will then be passed to one or more fully connected layers, where the model will make its inference. Furthermore, CNNs were used in background works 1 and 3 which are explained above. The structure is detailed below.

An input 256x256 image will first be zero-padded. This is to ensure good coverage of the edges of the image. The image will then be sent to a convolutional layer, where a filter will apply the convolutional operator over the image. The filter will be initialized randomly at first, and tuned during the training stage with backpropagation. Following this, a rectified linear unit (RELU) function will be applied to the feature maps to introduce non-linearity. The output of this will be sent to a pooling layer to reduce the size of feature maps. This combination of convolution layers, followed by the non-linear RELU function and then a pooling layer, would be repeated several times to extract many different feature maps, forming the feature learning stage.

This is followed by the classification stage, in which the output from the feature learning stage is flattened and sent to one or more fully connected layers. In this stage will consist of simple neurons followed by a RELU function. Since the project involves binary classification, the very last layer will apply a sigmoid function, resulting in the final probability of a tumor detected in the image. Refer to Figure 1 (below) for a visualization of the architecture.

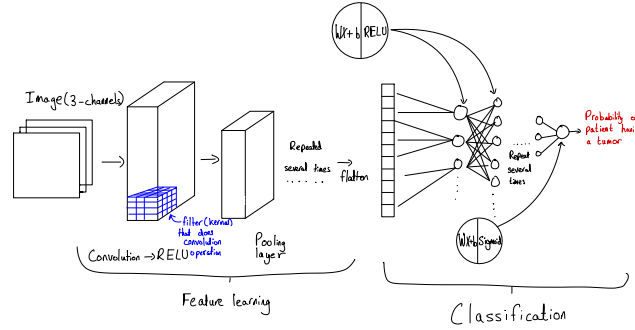


Figure 1: Illustration of the Model Architecture.

5 BASELINE MODEL

The proposed baseline model is a Support Vector Machine (SVM) to classify MRI images into tumor and non-tumorous categories. SVM has been widely used for brain tumour detection in medical images, demonstrating strong performance in previous studies (e.g., 97% accuracy for certain high glioma cases using T1 modality) as shown by (Tanneedi, 2017). SVM effectively handles complex classification tasks, particularly when using features extracted through the Histogram of Oriented Gradients (HOG) method.

This study assessed the SVM's performance using metrics such as accuracy, sensitivity, and specificity based on Receiver Operating Characteristics (ROC). For high glioma (a type of tumor) cases, SVM achieved a high accuracy and a true positive rate of 0.97, highlighting its strength in medical image classification. These results demonstrate the suitability and reliability of the SVM as a benchmark for comparison against our architecture.

A similar approach will be adopted by extracting features from MRI images using the Histogram of Gradients Method (HOG) and training an SVM classifier using the Radial Basis Function (RBF) kernel. The RBF kernel works by mapping the input features collected into a higher-dimensional

space. This is vital for capturing complex patterns in medical images that may not be linearly separable, such as varying tumor shapes, sizes, and textures. This baseline is suitable as a benchmark for the deep learning model due to its accuracy and precision.

6 ETHICAL CONSIDERATION

Since the model will be trained using medical images this causes a privacy concern since these images tend to contain sensitive information about patients. Furthermore, a lack of diversity in the data set can be problematic since the same type of brain tumor could look different in two patients with different characteristics such as race, ethnicity, and gender. Having a diverse dataset therefore ensures the model is fair and accurate for people of all backgrounds and traits.

There are many considerations in cases where the model produces inaccurate results. For example, if the majority of the errors are false positives, then this would lead to unnecessary stress for non-tumorous patients who are falsely classified as having a tumor. This could also lead to a waste of resources as false positives would need further testing. On the other hand, if the model overly reports false negatives, this spells disaster for people who have tumors but are undetected, which can lead to fatal consequences.

7 PROJECT PLAN

7.1 MEETINGS

The team will meet on Zoom on weekends (Saturdays or Sundays), to provide updates on the current progress of tasks and to work together in case some tasks require synchronous collaboration. The team will decide which day on the weekend to meet every week to allow for additional flexibility.

7.2 METHOD OF COMMUNICATION

The primary method of communication will be a texting group chat in iMessage.

7.3 CODE COLLABORATION

We will ensure that we will not overwrite each other's code by trying to divide each other's code into different files whenever possible and using git for version control.

7.4 TASK DIVISION TABLE

Table 1 below outlines the current and upcoming tasks in addition to assigned team member(s), and internal and external due dates for a given task.

Table 1: Task breakdown among team members, classified by category.

Category	Task	Assigned Member(s)	Internal Due Date	Due Date
Proposal	Project Idea Generation	All Members	09/29	10/04
	Introduction	Talha	10/02	10/04
	Illustration/Figure	Sadra	10/02	10/04
	Background and Related Work	Sadra	10/02	10/04
	Data Processing	Saleh	10/02	10/04
	Architecture	Talha	10/02	10/04
	Baseline Model	Taha	10/02	10/04
	Ethical Considerations	Talha	10/02	10/04
	Project Plan	Saleh	10/02	10/04
	Risk Register	Taha	10/02	10/04
	Link to GitHub/Collab Notebook	Saleh	10/02	10/04
	References	All Members	10/02	10/04
Progress Report	Brief Project Description	Saleh	10/24	11/01
	Data Collection	Saleh, Talha	10/12	11/01
	Baseline Model	Taha, Sadra	10/26	11/01
	NN Model Training	All Members	10/26	11/01
	Individual Contributions and Responsibilities	All Members	10/28	11/01
	Notable Contribution, Data Processing Write Up	Saleh, Talha	10/28	11/01
	Notable Contribution, Baseline Model Section Write Up	Taha, Sadra	10/28	11/01
	Notable Contribution, Primary Model Section Write Up	All Members	10/28	11/01

8 RISK REGISTER

1. **Team member drops course:** If a team member decides to drop the course, they must inform the team at least one week in advance. This notice will allow us to redistribute the responsibilities and assess the impact on our project timeline. Furthermore, we will hold a team meeting to discuss task reassignments and deadlines.
2. **Team member fails to meet internal deadline:** If a team member fails to meet an internal deadline, we will hold a team meeting to discuss any challenges they are facing and provide support if needed. Furthermore, we will set clear expectations and establish a timeline with regular check-ins to monitor progress to mitigate the risk. We will also encourage open communication, allowing team members to voice any difficulties early on.
3. **The dataset might contain poor-quality MRI scans or have class imbalance:** We will apply data augmentation techniques to diversify our dataset, thereby improving the model's ability to generalize unseen data. To address class imbalance, we will oversample the minority class and use weighted loss functions during model training.
4. **The model may overfit the training data:** We will implement several techniques designed to improve generalization. This includes employing regularization methods such as L2, Dropout, and Early Stopping which help prevent the model from becoming too tailored to the training data. Additionally, we can also incorporate non-regularization methods such as Mini-batch gradient descent and batch normalization to enhance generalization.

9 LINK TO GITHUB REPOSITORY

<https://github.com/SalehMousavi/TumorClassifier>

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